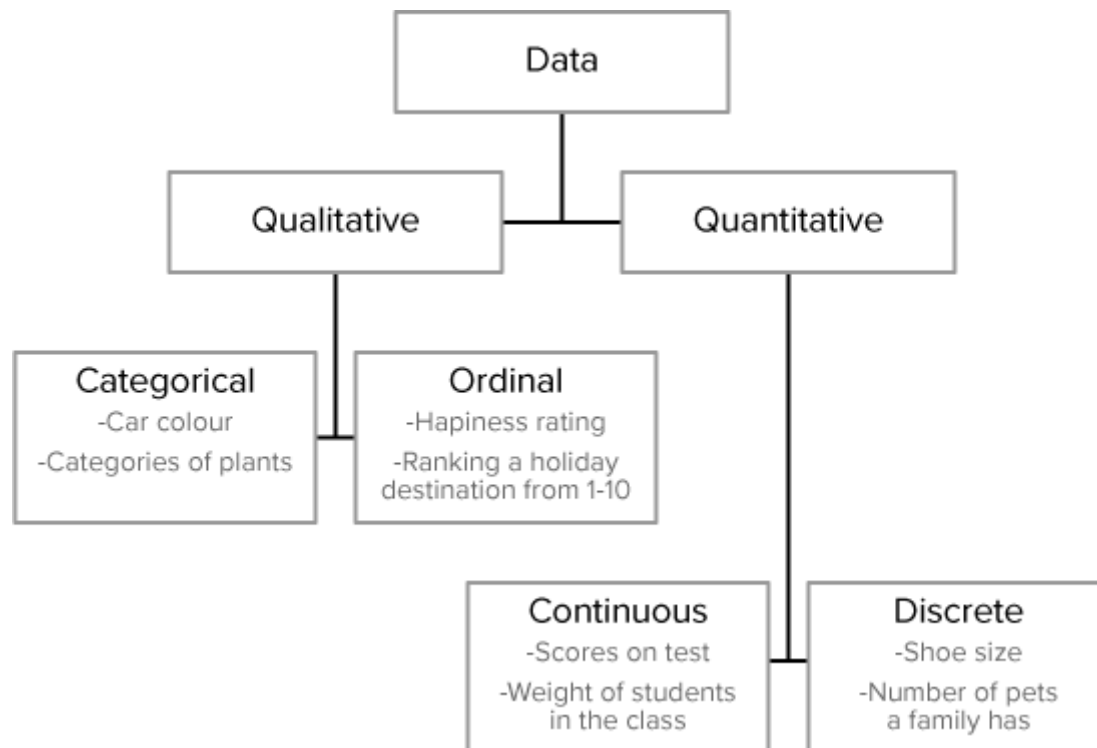


## Chapter Two: Data Collection and Presentation

The word data is plural, the singular being 'datum'. Dictionary meaning of the word datum is 'fact' and therefore in plural the word data signifies more than one fact. In a wider sense, the term data denotes evidence or facts describing a group or a situation. However, in a practical sense, the statistical term is generally used for numerical facts such as measures of heights, test scores on achievements, etc.

### Classification of Data

Data can be classified in the following ways:



[Figure 1]

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Data can also be classified as primary or secondary.

**Primary data** is data that is collected by a researcher from first-hand sources, using methods like surveys, interviews or experiments. It is collected with the research project in mind, directly from primary sources.

Whereas, **secondary data** is data gathered from studies, surveys or experiments that have been run by other people or for other research.

Data can also be classified as primary or secondary.

Methods of obtaining primary data:

- 1.Observation method**
- 2.Interview method**
- 3. Questionnaire metod**
- 4. Shedule method**

### **Observation Method**

The observation method is the most commonly used method specially in studies relating to behavioural sciences. In a way we all observe things around us, but this sort of observation is not scientific observation. Observation becomes a scientific tool and the method of data collection for the researcher, when it serves a formulated research purpose, is systematically planned and recorded and is subjected to checks and controls on validity and reliability.

#### **Advantages :**

- 1.Subjective bias is eliminated, if observation is done accurately.
- 2.The information obtained under this method relates to what is currently happening; it is not complicated by either the past behaviour or future intentions or attitudes.
3. this method is independent of respondent

#### **Limitations.**

- 1 It is an expensive method.
2. The information provided by this method is very limited.
3. Sometimes unforeseen factors may interfere with the observational task.

### **Interview Method**

The interview method of collecting data involves presentation of oral-verbal stimuli and reply in terms of oral-verbal responses. This method can be used through personal interviews and, if possible, through telephone interviews.

Interview method can be classified in two types .

(a) *Personal interviews*: Personal interview method requires a person known as the interviewer asking questions generally in a face-to-face contact to the other person or persons.

## **Advantages :**

### 1.Accurate and Reliable Data.

The interviewer can clarify questions and ensure that the respondent understands them properly, leading to more accurate answers.

### 2.Opportunity for Clarification.

If a respondent does not understand a question, the interviewer can explain it immediately.

### 3.High Response Rate.

Since the interviewer meets respondents face-to-face, refusal rates are usually low compared to mailed questionnaires.

### 4.Observation of Non-verbal Cues

The interviewer can observe body language, facial expressions, and tone, which help in better interpretation of responses.

### 5.Suitable for Illiterate Respondents.

It is very useful in areas where respondents may not be able to read or write.

### 6.Flexibility.

The interviewer can modify the order of questions or probe deeper for more detailed information.

### 7.Collection of In-depth Information.

Personal interviews allow for detailed and comprehensive data collection.

## **Limitations.**

### 1.Costly.

It requires travel, time, and trained interviewers, making it expensive.

### 2.Time-Consuming.

Conducting face-to-face interviews takes a lot of time, especially for large samples.

### 3.Interviewer Bias.

The interviewer's tone, attitude, or personal views may influence the respondent's answers.

### 4.Limited Coverage.

It is difficult to cover a wide geographical area.

### 5.Lack of Anonymity.

Respondents may hesitate to provide honest answers on sensitive topics.

### 6.Training Required.

Skilled interviewers are needed to collect accurate data.

### 7.Possibility of False Responses.

Respondents may give socially desirable answers instead of true opinions.

(b) *Telephone interviews*: This method of collecting information consists in contacting respondents on telephone itself. It is not a very widely used method, but plays important part in industrial surveys, particularly in developed regions.

**Advantages :**

1. It is more flexible in comparison to mailing method.
2. It is faster than other methods i.e., a quick way of obtaining information.
3. It is cheaper than personal interviewing method; here the cost per response is relatively low.
4. Recall is easy; callbacks are simple and economical.
5. There is a higher rate of response than what we have in mailing method; the non-response is generally very low.
6. Replies can be recorded without causing embarrassment to respondents.
7. Interviewer can explain requirements more easily.
8. At times, access can be gained to respondents who otherwise cannot be contacted for one reason or the other.
9. No field staff is required.
10. Representative and wider distribution of sample is possible.

**Limitations.**

1. Little time is given to respondents for considered answers; interview period is not likely to exceed five minutes in most cases.
2. Surveys are restricted to respondents who have telephone facilities.
3. Extensive geographical coverage may get restricted by cost considerations.
4. It is not suitable for intensive surveys where comprehensive answers are required to various questions.
5. Possibility of the bias of the interviewer is relatively more.
6. Questions have to be short and to the point; probes are difficult to handle.

## COLLECTION OF DATA THROUGH QUESTIONNAIRES

This method of data collection is quite popular, particularly in case of big enquiries. It is being adopted by private individuals, research workers, private and public organisations and even by governments. In this method a questionnaire is sent (usually by post) to the persons concerned with a request to answer the questions and return the questionnaire.

**Advantages :**

1. Low Cost.

It is an economical method of data collection, especially when questionnaires are sent by mail or online.

2. Wide Coverage.

A large number of respondents can be covered within a short time, even if they are geographically dispersed.

3. Time Saving.

Since there is no need for face-to-face interviews, it saves the researcher's time.

4. No Interviewer Bias.

The absence of an interviewer reduces personal bias and influence on respondents' answers.

#### 5.Easy Data Analysis.

Structured questions make coding, tabulation, and statistical analysis simple and systematic.

#### 6.Convenience for Respondents.

Respondents can complete the questionnaire at their own pace and suitable time.

### **Limitations.**

#### 1.Low Response Rate.

Many respondents may not return the questionnaire, leading to insufficient data.

#### 2.Lack of Clarification.

If a respondent does not understand a question, there is no opportunity to explain it.

#### 3.Incomplete Answers

Respondents may skip questions or give partial answers.

#### 4.Possibility of False Information

Some respondents may provide incorrect or dishonest answers.

#### 5.Limited Depth of Information

It is difficult to obtain detailed or in-depth information through fixed questions.

#### 6.Not Suitable for Illiterate People

This method is not effective if respondents cannot read or write.

#### 7.No Control Over Respondents

The researcher cannot control who actually fills out the questionnaire.

#### 8.Delay in Data Collection

Collecting and receiving completed questionnaires may take a long time.

### **COLLECTION OF DATA THROUGH SCHEDULE:**

The schedule method is a technique of collecting primary data where the investigator visits the respondents, asks questions directly, and records their answers in a structured form.

### **Difference Between Questionnaire Method and Schedule Method:**

| Basis                          | Questionnaire Method                                           | Schedule Method                                                      |
|--------------------------------|----------------------------------------------------------------|----------------------------------------------------------------------|
| Meaning                        | A list of questions sent to respondents to fill by themselves. | A list of questions asked by an interviewer who records the answers. |
| Who fills the form?            | Respondent fills it.                                           | Researcher/enumerator fills it.                                      |
| Personal Contact               | No direct personal contact.                                    | Direct face-to-face contact.                                         |
| Suitable for Illiterate People | Not suitable.                                                  | Suitable.                                                            |

|                            |                     |                                                               |
|----------------------------|---------------------|---------------------------------------------------------------|
| Cost                       | Low cost.           | More expensive (needs trained enumerators).                   |
| Response Rate              | Usually low.        | Usually high.                                                 |
| Chance of Bias             | No interviewer bias | Possibility of interviewer bias.                              |
| Clarification of Questions | Not possible.       | Possible (interviewer can explain). Designing a Questionnaire |

Properties of a good questionnaire:

- i) **Covering letter**
- ii) **The number of questions should be as few as possible.**
- iii) **Questions should be logically arranged.**
- iv) **Questions should be short and simple.**
- v) **Questions of a sensitive nature should be avoided.**
- vi) **Instructions to the informants.**
- vii) **Footnotes**
- viii) **Questions should be capable of objective answer.**
- ix) **Answer to questions should not require calculations.**

The following points may be kept in mind while designing a questionnaire:

a. Covering letter

-  Enclosed short letter;
-  Enclosed short address;
-  Assure the respondents information will be confidential.

- b. The number of questions should be as few as possible;
- c. Question should be logically arranged;
- d. It should be short and simple;
- e. Sensitive question should be avoided;
- f. Give instructions to the informants;
- g. Footnotes;
- h. Question should be capable of objective answer;

Various types of question that may form of a questionnaire

- 1. Dichotomous question;
- 2. MCQ;

### 3. Free answer/ open ended questions.

#### **Online Method (Online Survey Method)**

##### Meaning

The Online Method is a data collection technique where information is gathered through the internet using tools like email surveys, Google Forms, social media, or research platforms.

##### **Features:**

Conducted via internet.

Uses digital forms or survey software.

Respondents answer using computers or smartphones.

Data is automatically stored and organized.

##### **Advantages:**

Low cost.

Fast data collection.

Wide geographical coverage.

Automatic data recording and analysis.

Easy to distribute.

##### **Disadvantages:**

Limited to internet users.

Low response rate in some cases.

Risk of fake or careless responses.

Less control over respondents.

##### Example

A researcher using Google Forms to collect students' opinions about online classes.

#### **Case Study Method**

##### Meaning

The Case Study Method is an in-depth study of a single individual, group, institution, event, or community over a period of time.

##### **Features:**

Detailed and intensive study.

Focuses on one or a few cases.

Uses multiple sources (interviews, observation, documents).

Qualitative in nature.

### **Advantages:**

Provides deep and detailed information.  
Helps understand complex situations.  
Useful for social and psychological research.  
Generates new ideas or hypotheses.

### **Disadvantages:**

Time-consuming.  
Not suitable for large populations.  
Difficult to generalize results.  
Possibility of researcher bias.

Example

A detailed study of one successful Islamic bank to understand its management system.

### **Secondary data.**

Secondary data is the data that is collected by another person or organization and is available in published or unpublished form.

### **Sources of Secondary Data:**

1. Published Sources
  - Government publications (e.g., census reports)
  - Books and journals
  - Newspapers and magazines
  - Research reports
  - International organizations (World Bank, IMF, UN reports)
  - Websites and online databases
2. Unpublished Sources
  - Company records
  - Office files
  - Personal diaries
  - Thesis and dissertations

### **Comparison between primary and secondary data :**

| Basis of Comparison | Primary Data                                                        | Secondary Data                                  |
|---------------------|---------------------------------------------------------------------|-------------------------------------------------|
| Meaning             | Data collected first-hand by the researcher for a specific purpose. | Data already collected and published by others. |
| Source              | Surveys, interviews, observation, experiments.                      | Books, journals, reports, census, websites.     |
| Originality         | Original and raw.                                                   | Not original (already processed).               |
| Purpose             | Collected for current research problem.                             | Collected for some other purpose.               |



|               |                                                     |                                                    |
|---------------|-----------------------------------------------------|----------------------------------------------------|
|               |                                                     |                                                    |
| Cost          | Expensive.                                          | Less expensive.                                    |
| Time Required | Time-consuming.                                     | Saves time.                                        |
| Accuracy      | More accurate and reliable (if properly collected). | May be less reliable depending on source.          |
| Control       | Researcher has full control over data collection.   | No control over how data was collected.            |
| Suitability   | Highly specific to research objectives.             | May not perfectly fit research objectives.         |
| Examples      | Questionnaire, personal interview, field survey.    | Government reports,census data, research articles. |